A fantastic title about minihalos that somehow grips a viewers attention for over 100 pages of detail

by

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Abstract

A truly brilliant abstract that captures the nuance of what I have done, without being overly detailed and dense so that a reader throws it out the window.

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People who helped me during my Thesis: Yvette, CASA people, sibling.

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Chapter 1

Introduction

When looking up into the sky on a clear night those without an interest in astrophysics might imagine many of the points in the sky as galaxies just like our own, floating alone in the deep void of space. Although true to an extent, it doesn't paint an accurate picture of the cosmos. In reality, many of these form galaxy clusters, with tens to thousands of individual galaxies becoming gravitationally bound to a single massive structure (somewhat analagous to the planets in our solar system, but on a vastly different scale).[1] Galaxy clusters are the largest gravitationally bound structures in the universe, with total masses of order $10^{14}-10^{15}\mathrm{M}_{\odot}$ and diameters larger than 1.5 million parsec.[2]

One might naturally expect that majority of the cluster mass to be from the innumerable galaxies held within, however, only a few percent of the total cluster mass comes this baryonic matter (stars, planets, etc). The remainder of the baryonic mass comes from a hot diffuse gas (10^7-10^8 K) spread throughout the cluster called the intracluster medium (ICM). This dilute gas ($\sim 10^3$ particles m^{-3}) is held together by the gravitational pull of the cluster, with the highest density in the core and sparsest at the edges (Meekins et al., 1971; Serlemitsos et al., 1977). However, this amount of visible baryonic matter only comprises 15% of the gravitational mass calculated from the dynamics of the cluster, a huge discrepancy that was a key piece of evidence for the existence of dark matter. Indeed, the rough consensus from decades of study on galaxy clusters has $\sim 3\%$ of the mass budget taken by galaxies, $\sim 12\%$ from the ICM, and a staggering $\sim 85\%$ from unobservable dark matter (Blumenthal et al., 1984; Vikhlinin et al., 2006; Pratt et al., 2019).

Long filaments of galaxies span across the space between clusters, with their own warm-hot intergalactic medium (WHIM) mimicking the role of the ICM. The WHIM is cooler and more diffuse than the ICM, with densities $\leq 10^2$ particles m^{-3} and temperatures 10^5-10^7 K. These

[1] Smaller clusters with only tens of galaxies are often known as groups.

[2] For those unaccustomed to astrophysical units, $\mathrm{M}_{\odot}$ is the solar mass and defined as $1\ \mathrm{M}_{\odot} \approx 10^{30}$ kg. A parsec is defined as $1\ \mathrm{pc} \approx 3 \times 10^{16}$ m. These unusual units are far more suited at the scale of the universe.

filaments are larger than clusters, though not gravitationally bound, and create a cosmic web in the sky with galaxy clusters located in the intersections of the web. This cosmic web is the largest structure in the universe, a vast network of dark matter that we can trace via the observable galaxies and clusters (Bond et al., 1996; Pimbblet, 2005) with ongoing work in detecting these filaments observationally (Tudorache et al., 2025). Clusters are understood to then form in part due to the accretion at these intersections of the WHIM and the cosmic web, alonside the more dominant mergers of smaller clusters. These highly energetic merger events create shocks and turbulence within the cluster environment, causing the ICM to heat.

Add references to merger stuff here.

An important distinction should be made when analysing clusters, between "disturbed" clusters undergoing merger events or still in a turbulent state from a recent merger, and "relaxed" clusters which have settled into an undisturbed state. Relaxed clusters favour theoretical studies due to their greater spherical symmetries, fewer substructures, and therefore, simpler modelling (Mantz et al., 2016). Many relaxed clusters have also been found to feature a "cool core", where the temperature at the centre of the cluster is significantly lower than the region outside the core. The high density of the ICM in the core causes a large outflux of X-ray emission, great enough to lower the cooling time below the expansion rate of the universe (known as Hubble time). In a closed system, this develops into a cooling flow causing the core temperature to drop and gas flows inwards (Peterson and Fabian, 2006; Hudson et al., 2010). Observational evidence supports this theory, however the magnitude of observed gas is much lower than theorised, implying the existence of a heating mechanism to counteract the cooling flow. One such source are radio galaxies which are active galactic nuclei (AGN) that emit radio synchrotron emission. This emission create jets and lobes that have been observed to interact with the ICM, distributing energy in "radio-mode" feedback (Fabian, 2012). The exact method of this heating is still disputed and undergoing research.

Within the high mass subset of relaxed cool core clusters ($M_{500} > 6 \times 10^{14}$ $M_\odot$)[3] recent works have shown that 80% contain radio minihalos (Giacintucci et al., 2017). Less common in lower mass clusters, and unobserved within non-cool cores, minihalos are diffuse radio sources with currently unknown origins. This radio emission is localised within the cool cores, with radii from 50 kpc to $0.2R_{500}$.[4] This emission cannot be directly formed by radio sources such as AGN, as there is no observed spatial or spectral connection at any resolution or frequencies studied thus far. Furthermore, the spatial extent of these halos is well beyond the expected extent for particles sourced from an AGN (e.g. Bagchi et al., 2002; Giacintucci et al., 2014). Understanding the origins and physics behind minihalos is an ongoing process, with viable production methods proposed, and a lack of observational evidence to differentiate between them.

[3]The value M_{500} refers to the total mass within a radius of R_{500} where the clusters average enclosed density is 500 times the critical density of the universe at the cluster redshift. The critical density is a cosmological quantity, that has varyied over the lifetime of the universe.

[4]The radius $0.2R_{500}$ is a standard metric to separate the core and the outer ICM. Within this radius, the cooling, AGN feedback, and similar processes become significant to the behaviour of the ICM.

Working towards improving our understanding of minihalos will be the focus of the thesis. **Improve the ending of this section.**

1.1 Observations of Galaxy Clusters

The question that should then be asked is, how *do* we know all of this information about galaxy clusters? How are we able to measure the temperature, density, pressures within these clusters without travelling the impossibly vast distances to get there? The answer to this question is simple, by observing the radiation emitted from these structures. Different processes within the cluster, and more specifically ICM, provide information to the physical environment. This radiation is emitted across the electromagnetic spectrum and we design telescopes to capture and relay different bands in order to identify different components of what we observe. The question then becomes, how do observations at different frequency bands inform us to the structure and processes occuring within a cluster?

1.1.1 Optical

The vast majority of optical light from clusters comes from stars within the component galaxies. Although incredibly informing about the stars and galaxies themselves, in the context of clusters, optical light has limited use. One usage is the ability to calculate the radial velcoity of cluster galaxies, which can then be used to inform stuff about mass. The other usage is to combine it with radio or X-ray data in order to determine what the source of emission is. For example does a radio burst have an optical counterpart? Then its likely a galaxy instead of some floating relativistic gas.

1.1.2 X-ray

The hot gas of the ICM forms a thin plasma, which has a degree of ionisation that depends on the temperature and density. We observe an X-ray spectrum from this gas with both continuum and line characteristics, with photon energies between 0.1 keV and 100 keV.[5]

Line emission occurs due to electron transitions between orbitals within an atom. The temperature and density of the ICM changes the measured rate of this line emission, however, this becomes limited at higher temperatures as the ionisation fraction increases. The distribution of line emissions can also be an indicator of abundance of elements such as C, N Ne, Mg, Si, S, Ar, Ca, and Fe. This is particularly evident at lower temperatures ($\lesssim 2$ keV) where the total emissions can be dominated by line emission (Schneider, 2014).

[5]X-ray emission is historically measured in kilo-electron volts (energy) rather than frequency or wavelength.

Continuum radiation is due to brehmsstrahlung, or free-free radiation, which is where electrons are deflected via Coulomb interaction with nearby nuclei. This deflection causes an acceleration induces a change in kinetic energy which is emitted as a photon. As this process is

1.1.3 Radio

1.1.4 Gamma

- Well, we can't just go there so we get to look at them really closely.
- How do we get data about clusters
- In increasing wave length, Optical, X-ray, Radio, Gamma.
- Some telescope + research that we can do with each of these.

It has long been known that some clusters exhibit diffuse radio synchrotron emission that cannot be directly associated with radio galaxies within the cluster. This synchrotron emission is produced by particles travelling at relativistic velocities through a magnetic field, emitting photons at radio energies characterised by their velocity and the magnetic field strength (Feretti et al., 2012). This emission can take many spatial arrangements, with the primary categories being radio halos, radio relics, and revived AGN sources. Radio halos are extended sources that trace the ICM distribution and operate at the scales of giant radio halos, matching the full ICM extent, and minihalos that are contained within a clusters cool core. Relics instead reflect particles that have been re-accelerated to relativistic energies by shock waves travelling through the ICM, such as the violent energies released during merger events. Revived AGN sources are radio plasma that has been emitted from and AGN and re-energised similarly to the radio relics.

sources for drse

A key question in relation to the radio halos is how they are produced. Relativistic particles have a characteristic lifetime due to the energy losses as travel through space, and therefore a limited diffusion length-scale in which we should be able to observe them. Within a weak magnetic field ($\sim 10^{-7}$ G) these distances are about 10 pc, which even accounting for bulk motion of the ICM and more favourable diffusion environments are well below the Mpc scales observed in many clusters (Bagchi et al., 2002). One could argue that the powerful magnetic fields supermassive black holes like M87* have and its 1.5 Mpc long jet of relativistic particles could meet the length-scale criteria, but these strong fields also corral the particles like a funnel, preventing a radial diffusion as observed. Two primary theories have been brought forward to explain this phenomenon, one being turbulence in the ICM causing "re-acceleration" of once relativistic particles (e.g. Brunetti et al., 2001; Gitti et al., 2002; ZuHone et al., 2013; Zandanel et al., 2014). The other is inelastic collisions between relativistic cosmic ray protons and thermal protons, termed "hadronic" or secondary production (e.g. Dennison, 1980; Dolag and Enßlin, 2000; Keshet and Loeb, 2010; ZuHone et al., 2015; Ignesti et al., 2020).

Citation for M87*

Both of these mechnaisms produce similar observable features, which makes disentangling the true source much more challenging. Investigating into spatial extents and spectral features of ginat radio halos has given the re-acceleration theory more credence than hadronic production, however, the debate is still ongoing for minihalos. Due to their faint nature and smaller size, observations require high sensitivity and resolution in order to accurately capture an image.

1.2 Galaxy Clusters

- What is a Galaxy Cluster (grav bound galaxies, in a bunch of hot gas)
- Generic information about galaxy clusters (yay statistics)
- Fun facts about galaxy clusters

1.3 Radio Emission

- This is what we are interested in, can gain information about clusters from this.
- Link synchrotron emission to the electron / gas properies of the cluster and how radio emission can inform us of what the cluster is doing.
- Common sources of relativistic electrons are the AGNs, lose energy over time.
- Structures larger than the lifetime of these jets, more diffuse (Halos).

1.4 Minihalos

,

- What is a minihalo (diffuse radio emission in the core of a cluster.)
- They are somewhat common, things about detections.
- Difficulty in distinguishing between production mechanisms.

1.5 Our goals

- We want to be able to measure minihalo flux for this subset of minihalos at high radio frequencies so that we can investigate if steepening is occuring.
- This is the layout of the paper and what each chapter will cover in pursuit of said goal.

Chapter 2

Background

This chapter would be about radio interferometry, how it works to some extent and the data that is retrieved from it. Then we follow up with more specific data about radio emission.

2.1 What is an interferometer

- Limits of dish size in resolution, causes us to explore different ideas.
- How does an interferometer work.
- Bunch of dishes being interfered, delay to tune positioning, uv plane to determine coverage.
- From all of this we retrieve amp + phase measurements.

2.2 Radio emission in more detail

2.2.1 Large Radio Halos

- Use similar physics but much larger scale.
- Some basics things about previous research in differentiating production.
- Gamma ray non-detection.
- General consensus is that its re-acceleration.

2.2.2 Minihalos

- These are much smaller, so therefore much harder to observe. Difference to large radio halos.

- Similarities between host clusters (CC), general properties of spectral index etc.

2.2.3 Hadronic Production

- What is the process for hadronic production
- How does this manifest spectrally and evolve. What should we expect observationally here based on simulation work.
- How does this manifest spatially and evolve, what should we see observationally.

2.2.4 Turbulent re-acceleration

- What is the process here.
- How manifest spectrally.
- How manifest spatially.

2.2.5 Previous minihalo studies

- Generally look at lower frequencies, or a singular study for a particular minihalo.
- Want to look at more minihalos in a more holistic manner.
- Some findings from previous papers e.g. RXJ1720, MS1455+2232 in basic detail. (more detail on the per-halo discussion).

Chapter 3

Calibration and Imaging

This would be how to calibrate and the calibration results for the datasets, plus any special considerations we took.

3.1 Our observations

- We are using VLA data in X-band, what does this mean and what are the specs for the VLA.
- These observations are of list minihalos and this setup, observation proposal.
- Original source of these minihalos is the G14 paper (among others).
- Table of minihalos + observing setup.

3.2 How to calibrate a dataset

- We have the VLA calibration pipeline to remove stress from our lives.
- We have a non-neglible amount of RFI to add stress to our lives.
- A variety of steps but first pain, what is RFI and what does it cause.

3.3 Flagging data

- Initial testing showed that the pipeline flagging steps didn't work well for our data.
- AOFLAGGER worked much better and has a mostly out-of-box VLA script.
- How does AOFLAGGER work

- Integrating into a pipeline step to still use the pipeline

3.4 Calibration pipeline

- Work through each calibration step explaining what it does and why it is important.
- Final preparation for imaging is averaging down to further reduce noise.
- Notes about general trends from the calibration, with RFI in 10-12 GHz D config being bad, much of it being flagged.

3.5 Special mentions

- Either here or during following imaging section talk about the 2-3 datasets we processed slightly differently due to bad fits.
- Describe the fitting process for the custom bandpass.

3.6 Imaging

- When we make a raw image from the data, we still have a lot of artefacts from the point spread function, which comes from the beam.
- This means we need a more advanced imaging process to remove this to get a clean image. Implemented in a few different ways, we choose tclean in CASA.
- Tclean works by modelling the PSF, creating an empty model, find brightest point in residuals, model that as real, subtract model from residuals, repeat 20K times.
- This works mostly well, but we can improve results using self calibration.

3.6.1 Self calibration

- The phase calibration is the most iffy part for the main calibration, so we can make a test image and check the model seems reasonable.
- Then use that model to make small phase corrections to our dataset. Then repeat that a few times and these corrections should converge since it is circular.
- This is the final calibration steps that we do, phase corrections 20 degrees $\rightarrow$ 1 degree after a few rounds.

3.6.2 Final cleaning

- Between self calibrations this we also check the datasets and do some final manual cleaning.
- Clipping the datasets to a sensible amp range, removing spikes, SPWs where there is small amounts of data left.
- Overview of the `assess_spw` function used for this.

Chapter 4

How to Analysis

This chapter starts with an overview on imaging and how that is performed. Then we talk a bit about how we actually want to measure the flux of the minihalos. Could combine with actual results.

4.1 Final datasets

- Could have a table here talking about each minihalo, how much data is remaining, spws that are flagged out, some sort of visual.
- Maybe bar graph of spws, line or point graph for minihalo C/D (either total or differentiated by style) normalised to [0, 1]

4.2 Calibration Error

- Calibration error exists, and sources can vary in their flux amount.
- We can try and constrain the error and variance by comparing the secondary calibrator models for D config and the point source fluxes for the images.
- Fluxes for secondaries can vary a lot over a year (100%?) maybe, but lower per month. Can then compare to differences in sources in the image.
- If a trend is observed, likely calibration error. If no trend or close models, constrain calibration error and can see the variance in AGN etc.
- Can retrieve the point source flux by cutting out diffuse sources (uvrange) and by masking to a tight border.

4.3 Calculating the minihalo flux

- We can do this in a variety of ways, but we don't want the AGN contamination so we should subtract this out.
- Do this per dataset to account for variation over time, incorrect subtractions can result in negative bowls in emission, or remaining flux.
- Other problematic sources are dealt with the same way.
- Image the subtracted data and then we create a contour for significant emission (3 sigma) from the residual RMS.
- Calculate total flux in the region.
- Then we calculate the error using this formula from G14 modified for the AGN error since we think its lower.

Chapter 5

Results

- Sections for each minihalo with a brief explanation of imaging process, custom things we had to do to get results.
- Number of self calibration, sources subtracted, difficulties in imaging certain aspects.
- Discussion on the calibration error and variance in the observed point sources.
- Includes an image of the uvrange data showing point sources, image of the minihalo with contours, and anything else we desire.
- Includes discussion compared to previous results or measured values (likely G14) and any caveats or concerns.

Chapter 6

Discussion

- General findings from the above
- Any downsides caveats for the whole dataset
- Compare general results to things like G14 / G19 and how comparable these values are. Both literally and if valid comparison.

Chapter 7

Conclusion

- I'm sure I will have some conclusions here.

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